

Ethan Yang

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Education

University of Toronto

Expected Graduation: May 2030

Bachelor of Applied Science in Computer Engineering

Toronto, Ontario

- **GPA: 3.61**
- Relevant Coursework: Software Design and Communication, Introduction to Machine Learning, Operating Systems, Computer Architecture, Algorithms and Data Structures, Software Engineering

Work Experience

Sun Life Financial

May 2026 – Present

API Development Intern

Toronto, Ontario

- Cut new **Kafka Streams** service setup time from **2 days to under 3 hours** for **8 engineering teams** by architecting a **Java**-based code generation platform that guides technical users through a structured intake process and automates **API** scaffolding
- Reduced average documentation lookup time from **10 minutes to under 2 minutes** by writing **Python** and **Node.js** scripts to migrate **1,200+** legacy documentation pages into **Markdown**, enabling downstream **LLM** ingestion and search

MannLab

January 2026 – Present

Undergraduate Researcher

Toronto, Ontario

- Uncovered unsafe behavior in AI adaptation methods across **9,144 EEG data samples** by auditing **8 machine learning techniques** and building a **3-step diagnostic tool** in **PyTorch** to trace model failures back to their root cause
- Co-authored **3 research papers** on **EEG biofeedback** and consumer BCI safety (under review at **IEEE ISMAR** and **NeurIPS**) by conducting wearable computing and XR research under Professor Steve Mann
- Built an **XR** game in **Unity** designed to cut cocaine cravings by **30%** by using real-time **EEG** brain signal data to adjust gameplay as a drug-free alternative to cold-exposure craving therapy

Robotics for Space Exploration

September 2025 – April 2026

Software Developer

Toronto, Ontario

- Achieved **93.96%** on the System Acceptance Review at the University Rover Challenge (**20+ points** above cutoff) by building a **PyQt** rover control GUI integrated with **ROS2** for remote navigation and arm control
- Cut unsafe arm command errors by **40%** by developing and unit testing **ROS2** communication safety nodes in **Python** with fault-tolerant inter-process communication

Projects

Finance Digitization | [Source Code](#)

Next.js | TypeScript | AWS | PostgreSQL

- Automated cheque processing for **300+ UofT EngSoc users** and **100+ UofT Engineering Clubs** via **OCR**-based receipt parsing, reducing approval time from **3 days to 1 hour** with a dual e-signature workflow

Sinatra | [Source Code](#) | [Demo](#)

React | TypeScript | FastAPI | Python

- Launched a browser-based **DAW** converting vocals to **MIDI** via ML transcription, with AI chord generation and **Three.js** 3D video export; attracted a **40+ user waitlist**
- Eliminated install friction by engineering an end-to-end audio pipeline from mic input to multi-track **MIDI** rendering, clip-based editing, and **FFmpeg** WAV/MP4 export running entirely in-browser

HATSEYE - DeltaHacks 12 Finalist | [Source Code](#) | [Demo](#)

Python | Flask | OpenCV | Arduino

- Placed **top 6 of 144 teams** by building a voice-activated AI vision assistant for the visually impaired with real-time camera analysis and a dual-pipeline architecture: vision responses delivered via TTS while ultrasonic sensors drive haptic feedback in parallel

Lock Block - HackHive 2026 Winner | [Source Code](#) | [Demo](#)

Python | Flask | OpenCV | Arduino

- Won HackHive 2026 by building a smart lock using **OpenCV** face detection and recognition models, with **Ed25519** cryptographic wallet authentication for remote unlock and automated email alerts on unknown detection
- Prevented physical bypass attacks by engineering an **Arduino** servo-driven anti-pick locking mechanism, communicating lock state changes via **PySerial** between the vision backend and microcontroller

Leadership

UofTHacks

March 2026 – Present

Event Logistics Executive

Toronto, Ontario

- Secured a sponsor partnership for UofTHacks 14 by coordinating outreach to external clubs and companies while managing event logistics, guest scheduling, and room allocations

Technical Skills

Languages: Python, Java, C, C++, TypeScript, JavaScript, SQL, HTML5, CSS

Libraries/Frameworks: React, Next.js, Node.js, FastAPI, Flask, PyTorch, OpenCV, Three.js, PyQt, Tailwind CSS, Unity

Developer Tools: Git, Docker, AWS (ECS, S3), Apache Kafka, Apache Flink, ROS2, Postman, Jira, Arduino, PySerial, Vite